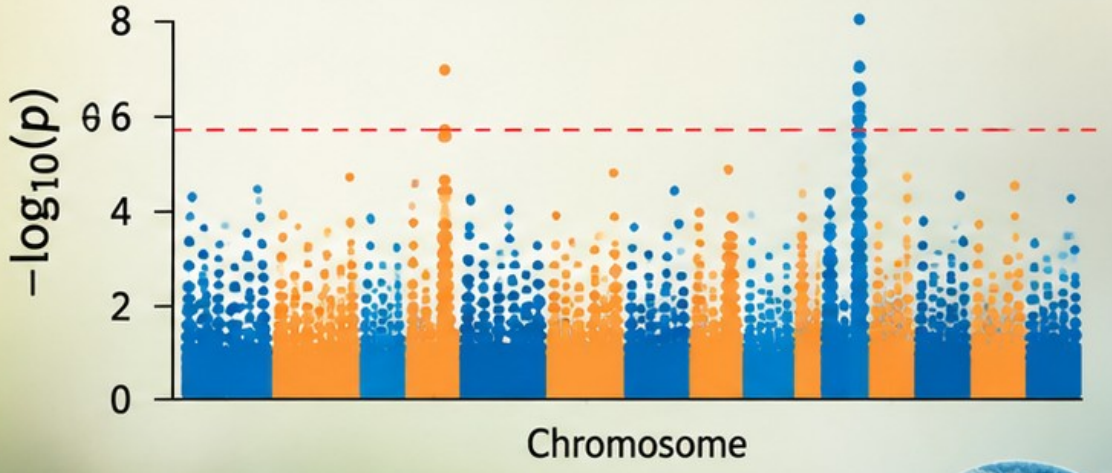
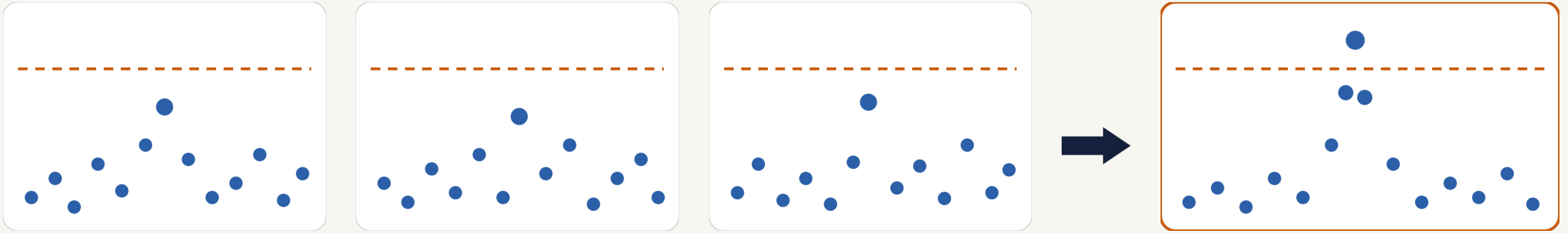
Genome-Wide Association Studies (GWAS) for Evolutionary Biology

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Why combine GWAS studies?



Study 1: too small | Study 2: too small | Study 3: too small | Meta-analysis: significant

More samples means more power, without sharing individual genotypes.

Schematic. Dashed line = genome-wide significance threshold.

What each study shares

Study	SNP	Effect / other allele	Effect β	Error SE	Samples N
Batch 1	chr3:1204	A / G	+0.80	0.35	10
Batch 2	chr3:1204	A / G	+1.10	0.25	10

β
Direction and size of the effect

SE
How certain the study is

Alleles
Align them first: flip swapped alleles

Example values are illustrative.

Fixed effect: weighted average

WEIGHT

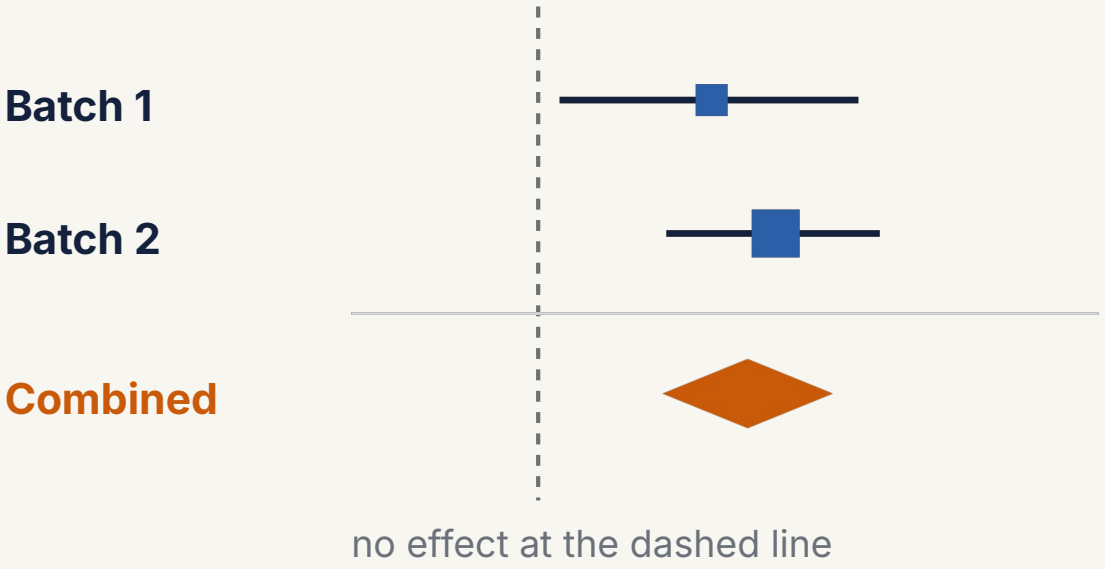
$$w = 1 / SE^2$$

COMBINED EFFECT

$$\beta = \Sigma(w \cdot \beta) / \Sigma w$$

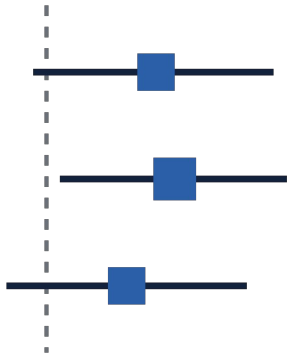
COMBINED ERROR

$$SE = 1 / \sqrt{\Sigma w}$$



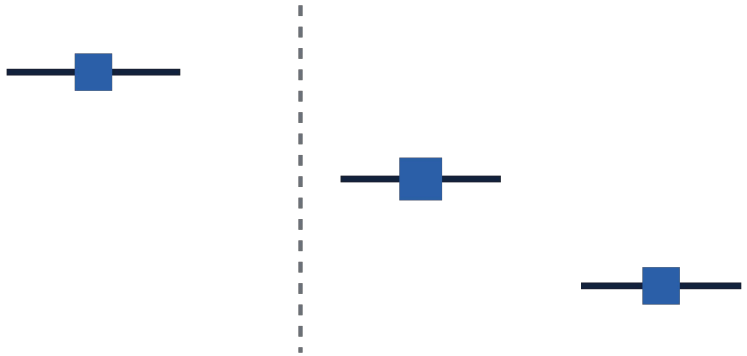
Formulae: Willer et al. 2010, METAL, Table 1. Plot is a schematic.

Do the studies agree?



Consistent

Intervals overlap. Low Q, I^2 near 0%.



Heterogeneous

Intervals do not overlap. High Q, high I^2.

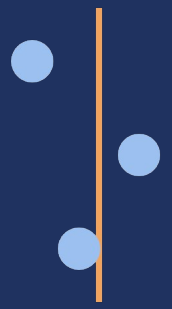
Q tests for disagreement. **I^2** is he percentage of the variation between studies' estimates that comes from real differences between the studies rather than from chance.

What I^2 means: studies never give exactly the same estimate, because each has sampling noise from its finite sample size. I^2 asks how much of the spread between estimates is more than that noise would produce.

Q measures the weighted spread of the study estimates around the combined estimate

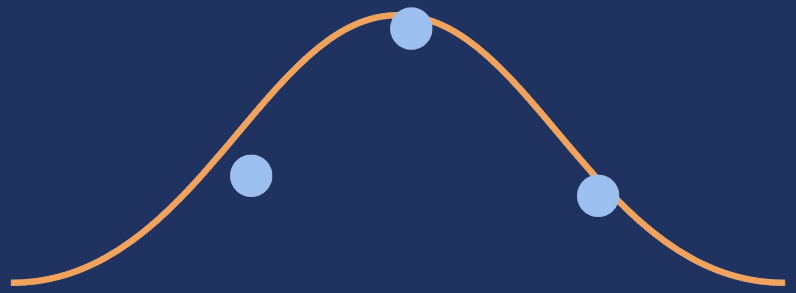
Cochran's Q and I^2: reported by both GWAMA and METAL. Plots are schematic.

Random effects: allow real differences



Fixed

One true effect. Differences are chance only.



Random

Each study has its own true effect. τ^2 = spread.

Weights become $1 / (SE^2 + \tau^2)$. Result: wider, more cautious intervals.

DerSimonian–Laird random effects, available in GWAMA (--random). Plots are schematic.

Two tools, one idea

GWAMA

- Fixed and random effects
- Q, I^2 and effect directions
- Needs β and SE per study

METAL

- STDERR: weighted β (as left)
- SAMPLESIZE: weighted z-scores
- Fast and scriptable, with Q and I^2

Both use the same input, one row per SNP per study, and both can apply genomic control.

Mägi & Morris 2010 (GWAMA); Willer, Li & Abecasis 2010 (METAL).

Our butterfly example

20 BAMs
m and i morphs

→

2 batches
10 each: 5 m + 5 i

→

ANGSD
doAsso per batch

→

GWAMA
and METAL

→

Results
Manhattan, QQ

Effect directions agree between batches?

Low heterogeneity (I^2) at the top SNPs?

Forest plot for each top SNP

Hypolimnas misippus; see the course notebook for the full pipeline.

References

GWAMA

Mägi R, Morris AP. GWAMA: software for genome-wide association meta-analysis. *BMC Bioinformatics* 2010;11:288.

METAL

Willer CJ, Li Y, Abecasis GR. METAL: fast and efficient meta-analysis of genomewide association scans. *Bioinformatics* 2010;26(17):2190–2191. doi:10.1093/bioinformatics/btq340

Figures in this deck are schematic drawings. See the original papers for their own figures and tables.